Topic: Autonomous smart irrigation system using ai

Harini.at.el 2024 published a paper in Smart irrigation systems revolutionizing farming.
The system makes decisions using sensors that monitor humidity,temperature, and environment to gather data at the appropriate times. With the use of artificial intelligence algorithms like machine learning models .

The advantage of this approach is that it will support local knowledge combined with technology application to enhance agricultural production and environmental protection.

The project's hardware includes the Arduino Uno controller as its core, the DHT11 sensor which measures temperature and humidity, as well as the soil humidity sensor; two solenoid valves control the water flow.

In the archive Survey of “Machine Learning countable Applications in Exactness Farming"examines the critical headways in advanced farming due to the integration of machine learning and fake insights. machine learning models in different perspectives of accuracy in agribusiness, including edit abdicate expectation, infection discovery, weed discovery, edit acknowledgment, trim quality evaluation, dribble water system, water quality administration, soil properties examination, climate determining, creature welfare, animal generation, collecting methods, and fertilizer suggestion.

Dataset It is a tree-structured classifier, where inside hubs speak to the highlights of a dataset, branches speak to the choice rules, and each leaf hub speaks to the outcome. In a choice tree, there are two hubs, which are the choice hub and leaf hub. The choices or the test are performed on the premise of the highlights of the given dataset. It is a graphical representation for getting all the conceivable arrangements for a problem or decision based on given conditions.

The results he data reveals a significant positive impact on water conservation and efficiency compared to conventional irrigation methods. based on real-time data, leading to reduced water wastage and optimized resource utilization. Furthermore, the implementation of the smart irrigation system demonstrates tangible benefits in terms of crop yield and quality.